

National University of Computer & Emerging Sciences, Karachi Campus



Project Report Of Applied Human Computer Interaction Redesigning Clash Royale with HCI Concepts

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1. Introduction and Game Selection

For this project, we selected the mobile game Clash Royale, a real-time multiplayer strategy game. The game is widely popular but contains several usability and user experience issues, especially for new users.

The reason for selecting this game is:

- Complex interface with multiple features
- High cognitive load for beginners
- Inconsistent navigation across screens
- Opportunity to improve readability and accessibility

The goal of this project is to analyze usability problems and redesign the interface using HCI principles to improve overall user experience.

2. Identified Usability Problems (Existing UI)

Based on the existing Clash Royale screens, the following usability issues were identified:

1. Cluttered Interface

- Too many elements on the home and deck screens
- Lack of spacing makes it difficult to focus

2. Poor Visual Hierarchy

- Important actions (e.g., Battle button) are not clearly prioritized
- UI elements compete for attention

3. Inconsistent Navigation

- Different layouts across screens (settings, deck, home)
- Users need time to relearn navigation

4. Small Touch Targets

- Buttons and icons are small, especially for mobile interaction
- Increases error rate

5. High Cognitive Load

- Too much information displayed at once (cards, stats, icons)
- Difficult for new users to understand quickly

6. Weak Feedback System

- Limited visual/system feedback for actions
- Users are unsure if actions are successful

3. User Analysis

To design an effective interface for Clash Royale, we identified the primary user groups:

Target Users

- 1. Casual Players**
 - Play occasionally
 - Prefer simple and quick navigation
 - Easily overwhelmed by complex UI
- 2. Competitive Players**
 - Play frequently
 - Need fast access to deck and battle features
 - Prefer efficiency over visuals
- 3. New Users (Beginners)**
 - No prior experience
 - Require guidance and clarity
 - Struggle with complex layouts

User Needs

- Easy navigation
- Clear instructions and feedback
- Fast access to core features (Battle, Deck)
- Readable interface

User Problems in Existing UI

- Confusing layout for beginners
- Too many elements on screen
- Lack of guidance
- Poor prioritization of actions

4. Heuristic Evaluation

We evaluated the original interface using Nielsen Norman Group's 10 Usability Heuristics.

1. Visibility of System Status

- Old UI: Limited feedback during actions
- New UI: Clear indicators (battle status, results)

2. Match Between System & Real World

- Old UI: Icons not always intuitive
- New UI: Better labels and clearer icons

3. User Control & Freedom

- Old UI: Difficult navigation
- New UI: Bottom navigation allows easy movement

4. Consistency & Standards

- Old UI: Inconsistent layouts
- New UI: Uniform design across all screens

5. Error Prevention

- Old UI: Small buttons → misclicks
- New UI: Larger buttons reduce errors

6. Recognition Rather Than Recall

- Old UI: Users must remember features
- New UI: Clearly visible options

7. Flexibility & Efficiency

- Old UI: Slow task flow
- New UI: Faster interactions

8. Aesthetic & Minimalist Design

- Old UI: Cluttered

- New UI: Clean and minimal

9. Help Users Recognize Errors

- Old UI: Weak feedback
- New UI: Clear results and responses

10. Help & Documentation

- Old UI: Limited guidance
- New UI: More structured layout helps understanding

5. Working of the Redesigned System

The redesigned interface of Clash Royale was developed by restructuring user interaction flow, improving navigation, and simplifying task execution. The system focuses on efficient user journeys and reduced cognitive effort.

5.1 Overall Workflow

The redesigned system follows a simple and consistent interaction flow:

Home Screen → Navigation Bar → Selected Feature → Action → Feedback (Result)

Steps:

1. User lands on Main Menu (Home Screen)
2. User selects an option using bottom navigation bar
3. User performs an action (Battle, Deck Edit, Settings)
4. System provides instant visual feedback
5. User returns or continues smoothly

5.2 Main Menu Working

- The Home Screen acts as the central hub
- A large “Battle” button is placed in the center for quick access
- Secondary options (missions, rewards) are placed below

Improvement:

- Reduces time to start a game
- Prioritizes primary action

5.3 Navigation System

- A **fixed bottom navigation bar** is used across all screens
- Sections include:
 - Home
 - Deck
 - Shop
 - Social

Working:

- Users can switch screens instantly without going back
- Navigation remains consistent throughout the app

Improvement:

- Eliminates confusion
- Improves user control and freedom

5.4 Gameplay HUD Working

- During battle, the interface displays:
 - Player status (top)
 - Arena view (center)
 - Cards and elixir (bottom)

Working:

- User selects a card → drags/drops on arena
- Elixir updates dynamically
- Real-time feedback is shown

Improvement:

- Cleaner layout improves focus
- Better spacing reduces errors

5.5 Deck Management Working

- Cards are displayed in a **grid layout**
- Users can:
 - Select cards

- Replace cards
- View stats

Working:

- Tap a card → view options → update deck

Improvement:

- Organized layout makes selection faster
- Reduces confusion compared to old cluttered UI

5.6 Settings Screen Working

- Settings are divided into **clear categories**
- Options use toggle switches and buttons

Working:

- User selects option → immediate system response

Improvement:

- Faster access to settings
- Better readability

5.7 Result Screen Working

- After battle, system shows:
 - Win/Defeat status
 - Rewards
 - Options (Play Again / Exit)

Working:

- User receives instant feedback
- Can quickly start next match

Improvement:

- Strong visual feedback
- Encourages continued engagement

6. Redesigned UI Overview

The redesigned interface includes the following screens:

1. Main Menu (Home Screen)
2. Gameplay HUD
3. Deck/Collection Screen
4. Settings Screen
5. Result Screen
6. Shop & Social Screens (additional improvements)

Your redesign uses:

- Dark theme with high contrast
- Simplified layout
- Clear navigation bar
- Consistent design across all screens

7. Application of HCI Principles

1. Consistency

- Same color scheme (dark blue + orange highlights)
- Uniform button styles and icons
- Consistent bottom navigation bar

2. Visual Hierarchy

- Primary action (Battle button) is large and centered
- Important sections are clearly separated
- Use of size, color, and spacing to guide attention

3. Cognitive Load Reduction

- Reduced clutter by grouping related elements
- Removed unnecessary visual noise
- Simplified card layout

4. Feedback & System Status

- Clear indicators for actions (e.g., battle status, rewards)
- Result screen provides clear win/loss feedback

5. Usability & Accessibility

- Larger buttons for easier interaction
- Better contrast for readability
- Cleaner typography

6. User Control & Freedom

- Easy navigation via bottom menu
- Clear back/forward actions
- Logical screen transitions

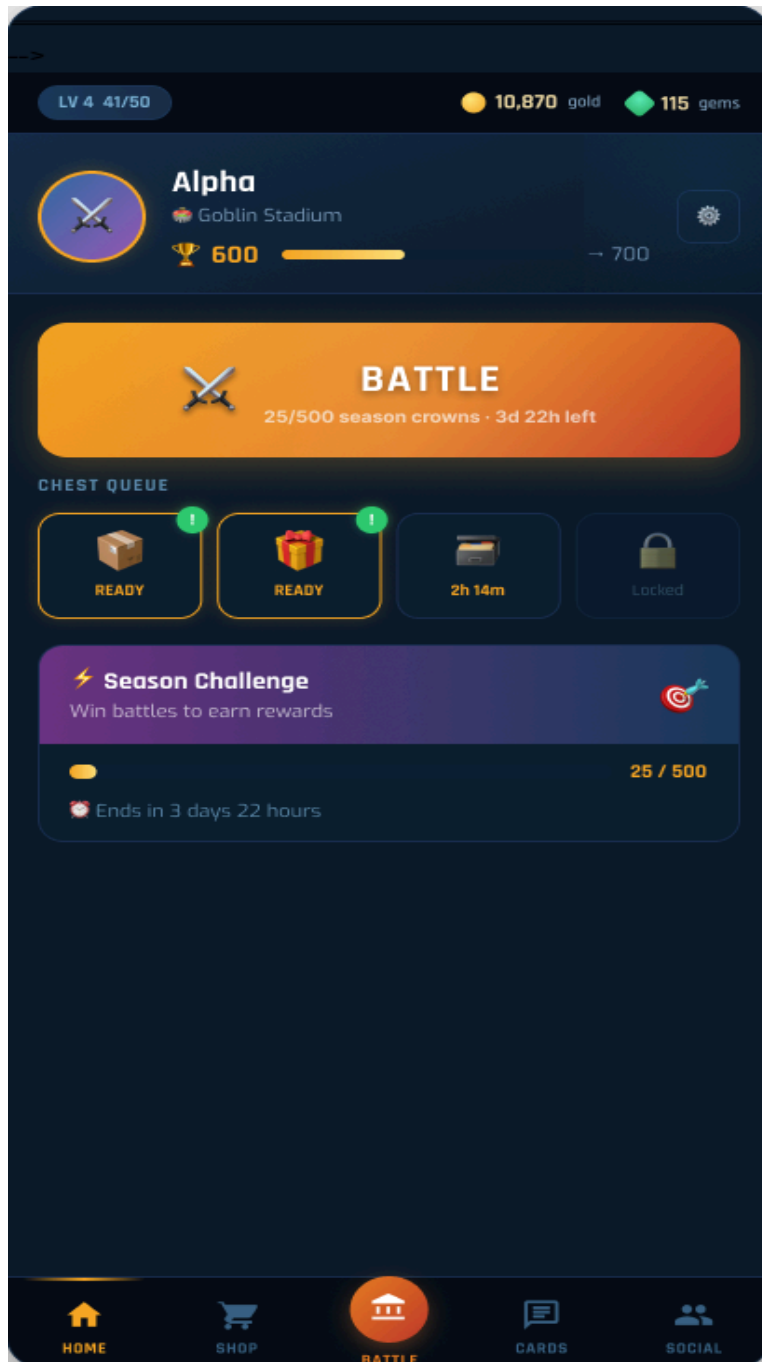
8. Screen-by-Screen Comparison (Before vs After)

1. Main Menu



Before:

- Cluttered with multiple icons and elements
- No clear focus



After:

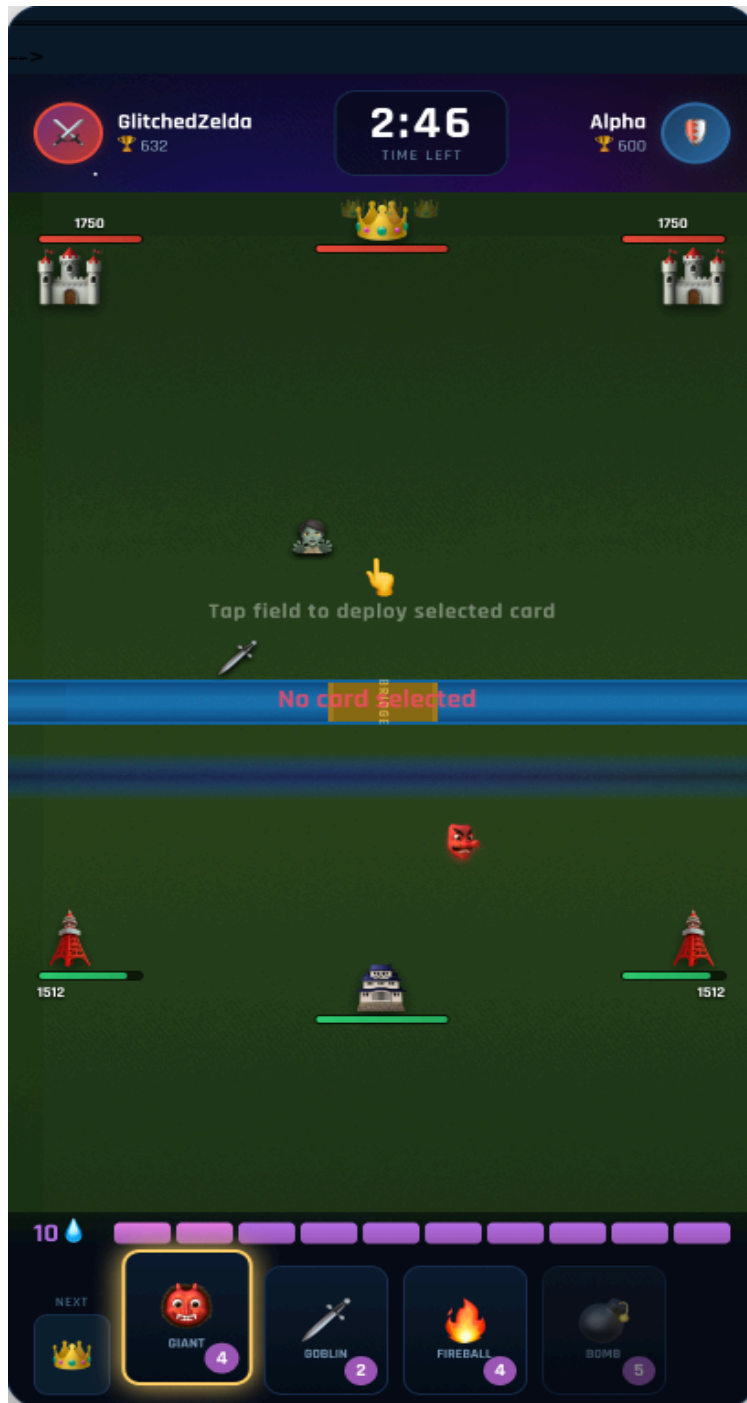
- Clean layout
- Large, centered Battle button
- Organized sections (missions, rewards)

2. Gameplay HUD



Before:

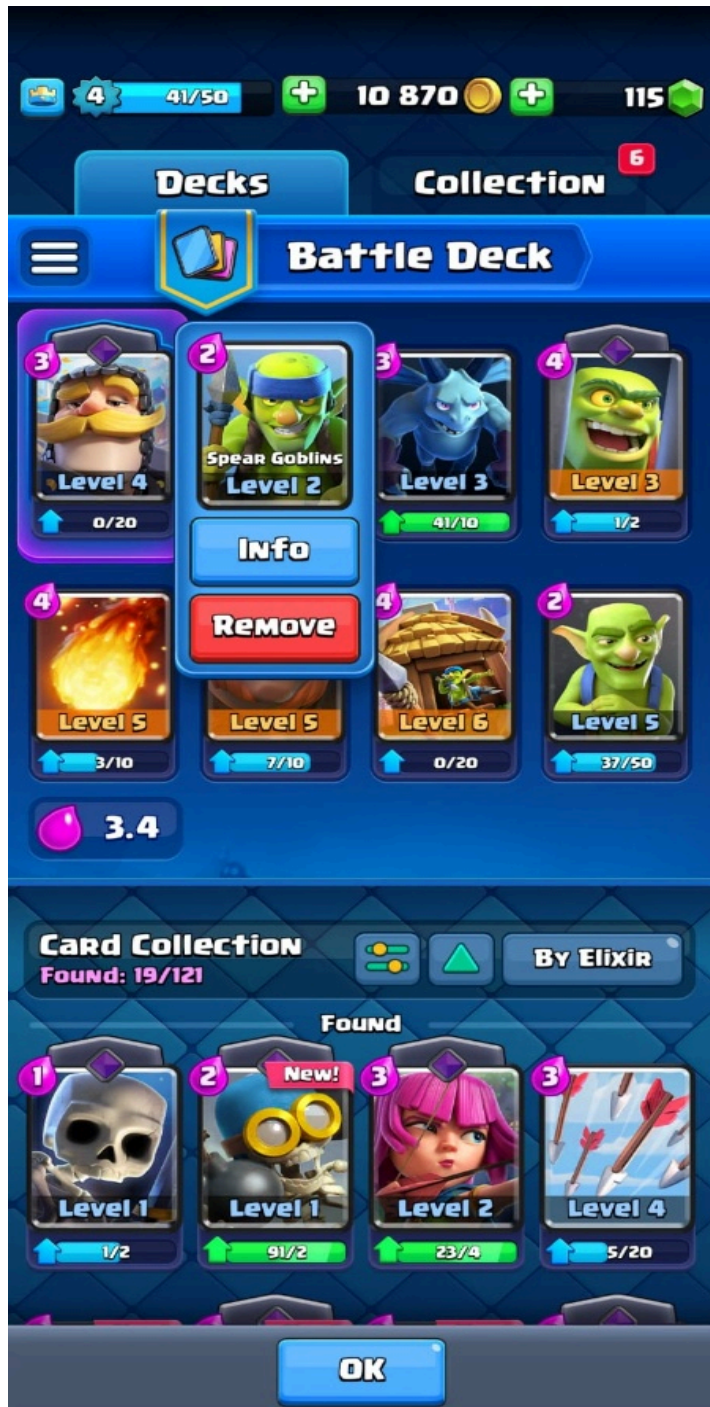
- Minimal guidance
- Weak visibility of controls



After:

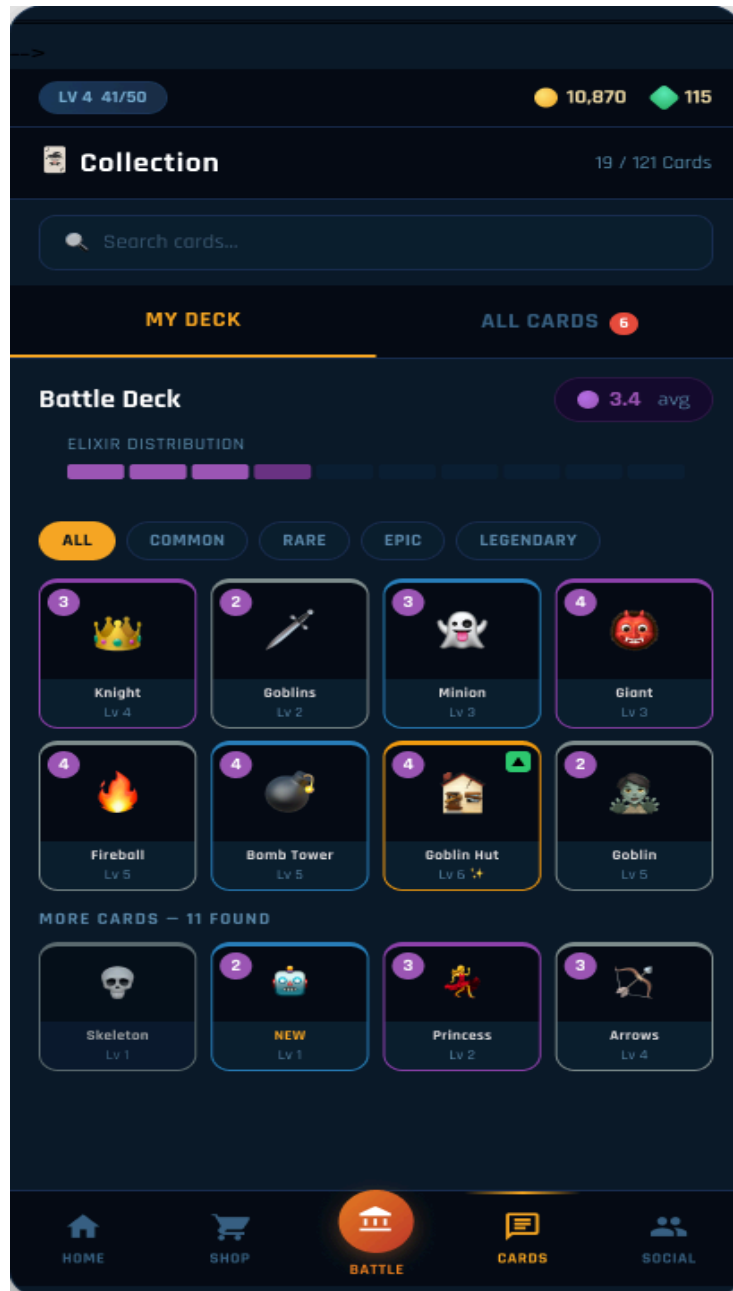
- Cleaner battlefield UI
- Better spacing for cards
- Improved visibility of elixir and actions

3. Deck / Collection Screen



Before:

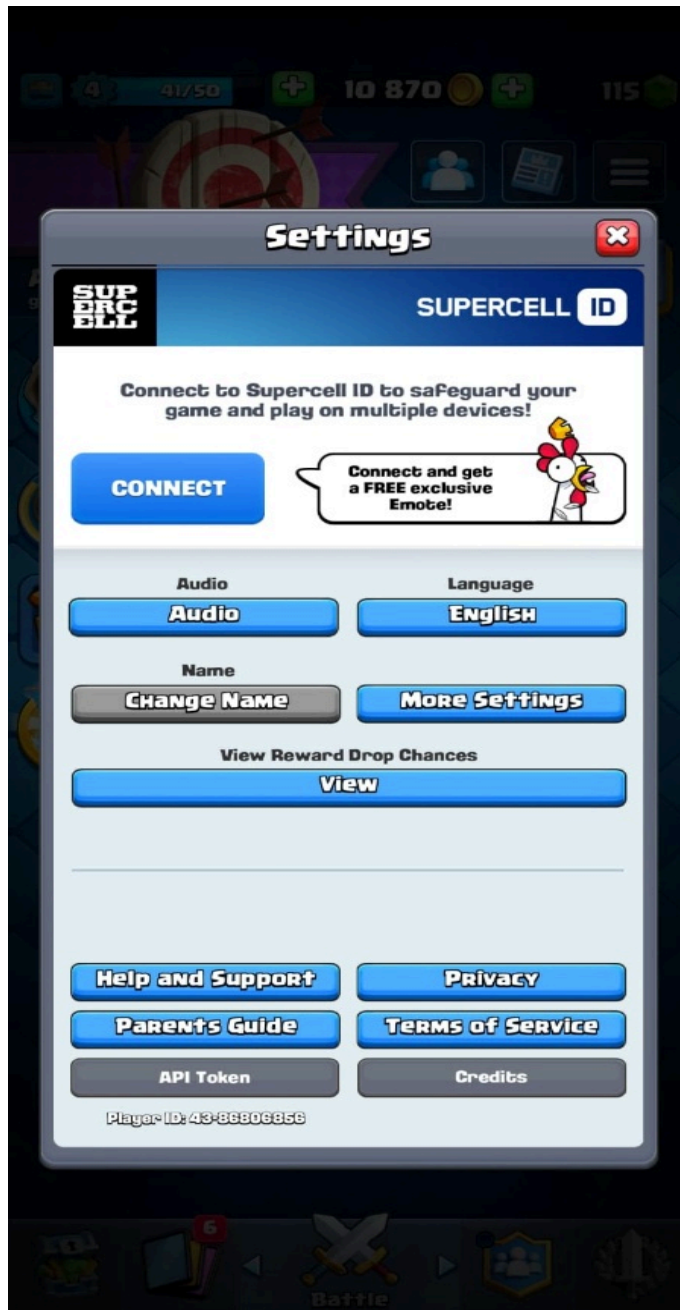
- Crowded card layout
- Hard to scan



After:

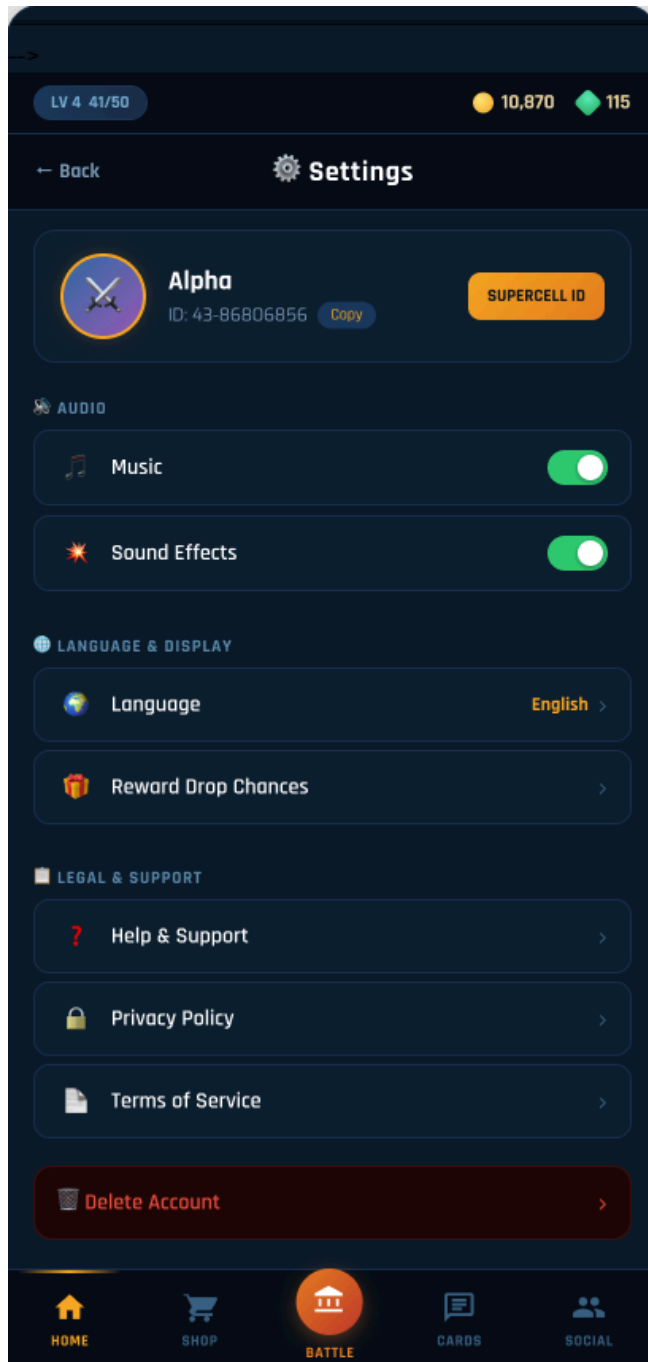
- Grid-based structured layout
- Clear card separation
- Easier selection and management

4. Settings Screen



Before:

- Basic layout with limited structure
- Poor grouping



After:

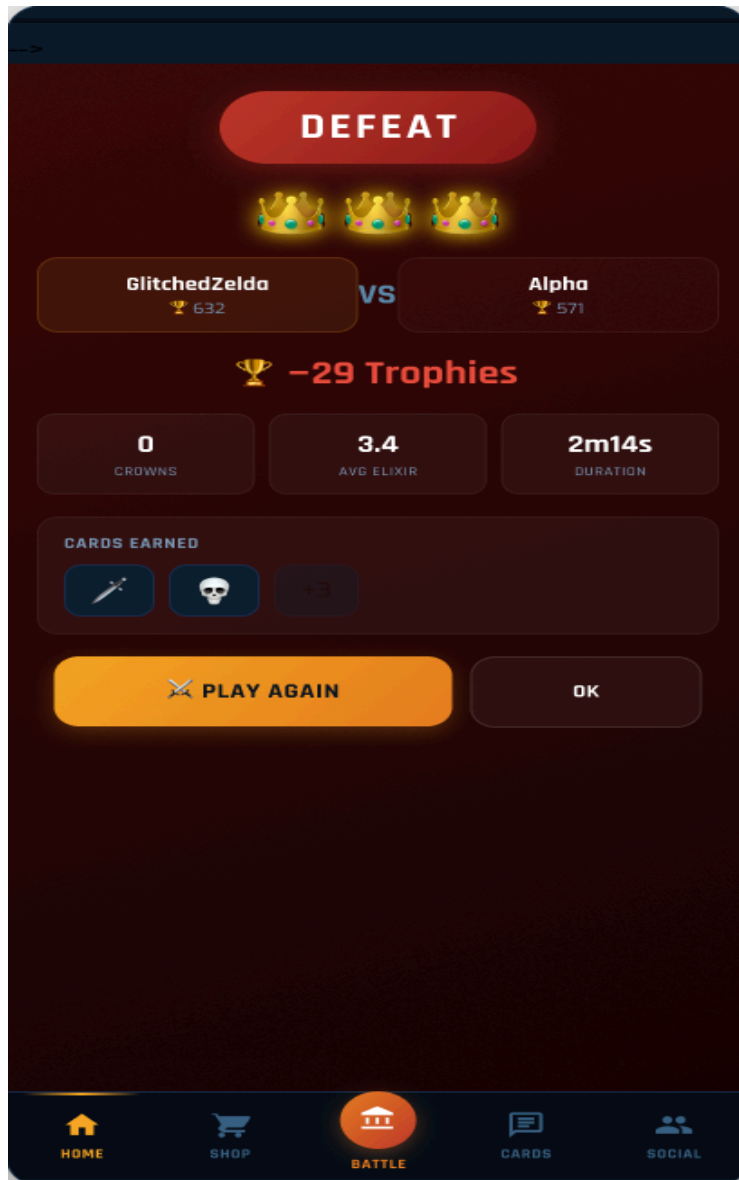
- Categorized settings
- Toggle switches for clarity
- Better readability and organization

5. Result Screen



Before:

- Basic feedback
- Limited visual impact



After:

- Strong visual feedback (Win/Defeat emphasis)
- Clear stats and rewards
- Improved engagement

6. Additional Screens (Shop & Social)

Redesigned:

LV 4 41/50

10,870 115



Shop

Refreshes daily

14:32:08

FEATURED

GOLD OFFERS

GEM PACKS

SEASON PASS



FEATURED DEAL - 40% OFF

Legendary Chest

Guaranteed Legendary card inside!



500 850



Buy Now

DAILY OFFERS



NEW

Silver Chest

4-6 cards inside

1,500 2,000



HOT

Epic Wildcard

Choose any Epic card

250



80 Gems

Starter gem pack

\$0.99 USD



Magic Items

Upgrade resources x5

3,000



SALE

Rare Wildcard

Choose any Rare card

800 1,200



Giant Chest

16-22 cards inside

180



HOME



SHOP



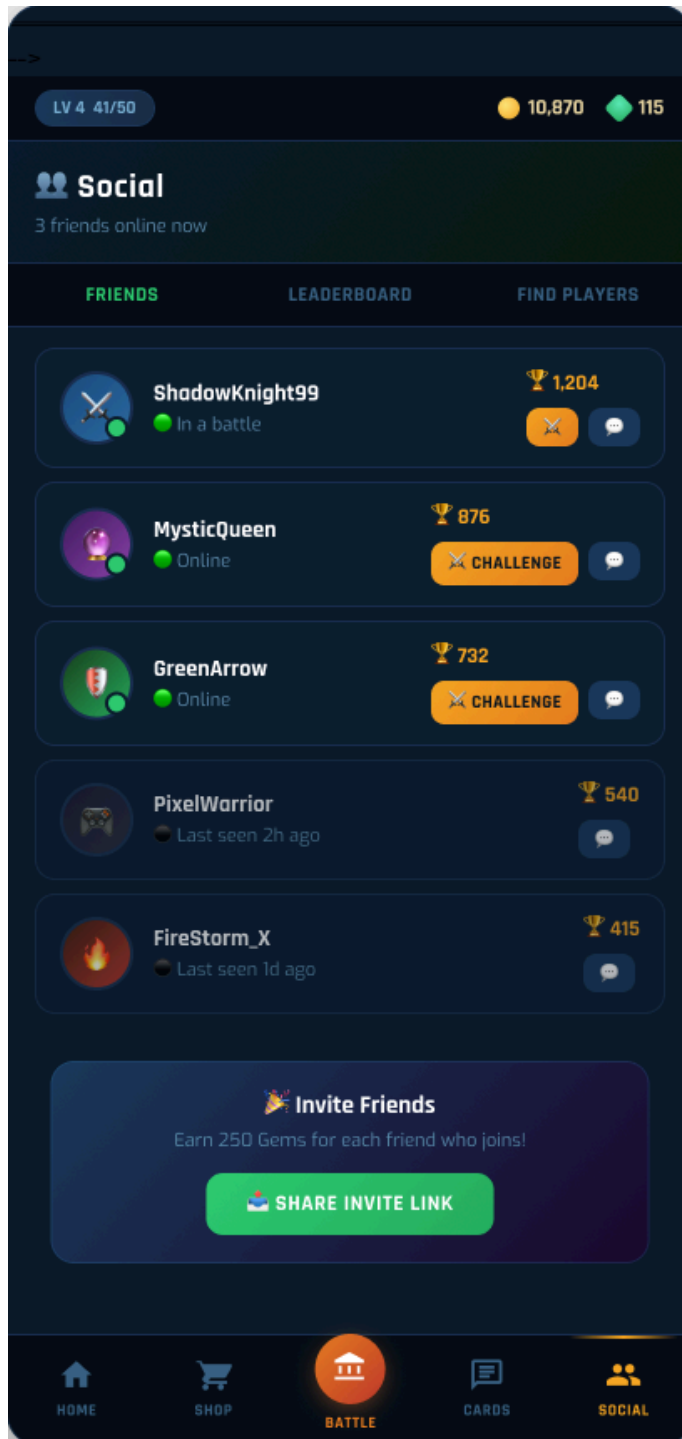
BATTLE



CARDS



SOCIAL



- Shop redesigned with card-style layout for offers
- Social screen structured with clear friend list and actions

9. Conclusion

The redesigned version of Clash Royale significantly improves usability and user experience by applying core HCI principles.

Key improvements include:

- Reduced cognitive load
- Improved navigation
- Better visual hierarchy
- Increased accessibility

The redesign demonstrates how thoughtful UI/UX design can enhance gameplay experience, especially for new users.